

Reading

PREPARING TO READ

A BUILD VOCABULARY The words in **blue** are used in the reading passage. Read the text below. Then write the correct form of each word next to its definition.

Many of the photos of Dutch artist Berndnaut Smilde are of clouds. However, these aren't normal clouds: they're clouds he made himself, inside of buildings. They don't last very long—some disappear after only 10 seconds—but they're definitely not **fake**: they're made of water vapor in the same way actual clouds are.

Smilde's work is part of an **ongoing** art series called *Nimbus*. To create each of his pieces, he first has to find a dramatic architectural environment. He then fills the space with water vapor using a simple spray bottle before using a smoke machine to release tiny particles into the moisture-rich air. These particles **trigger** the formation of small clouds by providing the water vapor with airborne surfaces on which to condense. Once a cloud starts to form, he quickly sculpts it into shape. And when it is finally the right shape and size—usually about six feet tall and ten feet wide—he photographs it in the few seconds it exists, before it dissipates into the air. The end result is **exquisite**—a magical shot of a perfect cloud in a beautiful, seemingly impossible indoor setting.

Smilde's work provides us with an **intimate** look at something that is typically only visible from a long distance away. He likes that his photos elicit strange and **ambiguous** feelings. However, the **impression** he most wants to evoke in his viewers is not the simple wonder of clouds appearing in unusual locations, but a feeling of transience—that his wispy creations exist only for a few seconds before they're gone. "I'm not interested in trying to create something that lasts forever," he says. The art that Smilde makes disappears almost as soon as it is created, rooting each of his pieces not just in a place, but in a specific moment in time.

1. _____ (v) to cause a process to begin
2. _____ (adj) vague or unclear
3. _____ (adj) not real, although it appears to be
4. _____ (adj) still happening
5. _____ (adj) extremely beautiful or appealing to the senses, in a way that is carefully crafted or delicate
6. _____ (adj) close, or deeply personal
7. _____ (n) a feeling one has that remains after meeting or encountering a person or thing

B BUILD VOCABULARY Complete the sentences below with the correct form of the words in **blue**. Use a dictionary to help you.

analytical (adj)

conservation (n)

envision (v)

exhibit (n)

foremost (adj)

imply (v)

interpretation (n)

theoretically (adv)

1. He didn't say he was angry directly, but he strongly _____ it.
2. The museum is full of _____ from all over the world.
3. The meaning of the movie isn't clear. It's open to _____.
4. Samuel doesn't make decisions based on how he feels. He prefers to take a more _____ approach.
5. It hasn't been proven experimentally, but _____, it's possible.
6. Sasha is an active supporter of several wildlife _____ projects.
7. Their work styles are so different. I can't _____ them working well together.
8. The guest of honor today is the world's _____ authority on microbiology.

C USE VOCABULARY Discuss these questions with a partner.

1. Have you ever been to an art gallery or display? Which **exhibits** stood out to you? Why?
2. What are some works of art that have left a strong **impression** on you? How did they make you feel?
3. Are some **interpretations** of art more valid than others? Why or why not? Whose interpretations should matter most?

D PREDICT Read the title of the reading passage. Then look at the photos. How do you think the art discussed in the passage makes an impact? Check your ideas as you read the passage.

Critical Thinking

One of Berndnaut Smilde's clouds inside the Shanghai Himalayas Museum



Making an Impact Through **Art**



The eyes of the African wild dog in this painting have the ability to draw viewers in.

Art has the power to inspire and effect change, as evidenced by these three examples.

ART FOR CONSERVATION

- A**  Stare into the eyes of *The Watcher*, British artist Sophie Green's portrait of an African wild dog, and you'll see there's something reflected. A triangular outline of a distant mountain perhaps, or maybe a termite mound on the savanna.¹ Something the animal is looking at, in any case, that draws and locks your own gaze. And by the time it does, you realize that the animal is actually now looking at you.
- B** The effect is striking: a strangely **intimate** moment with one of the planet's most beleaguered² mammals emerging from the shadows. But of course, it's not really an animal; just a very realistic painting of one.
- C** "That's always been my aim," says Green. "I want my artwork to be a window into another ecosystem. So people can feel they're face to face with the animal, rather than looking through a lens or at just another picture. Most people don't get that experience unless they go on a safari or an expedition. I kind of want my artwork to be that experience for them."
- D** Green's 14-piece exhibition—entitled *Impermanence: The Art of Conservation*—was initially **envisioned** to feature only polar animals, but Green says she quickly realized she was painting herself into a corner—conservation issues, human encroachment, and climate change affect animals all over the world, so she started to branch out. Hence images of balletic humpbacks, a great white shark, penguins, an African lion—and that African wild dog, amongst others.
- E** Green claims her art is not rooted in creativity. "I'm quite detail-focused, it's just my personality type. I would say I'm more **analytical**. I prefer something to be exact and precise."
- F** She says she paints in acrylic and adds that it's an unusual medium in hyper-realism. "It dries very quickly, but weirdly I prefer that," she says. "The way I build depth is with layer after layer after layer, and I do it quickly. It creates more of a depth perception."
- G** This lends itself to the hyper-real quality of her **exhibits**. "In watercolor, you work light to dark; you work dark to light in oil and acrylic. It's less a physical thing, more of a light perception. Light value on top of dark values kind of gives the **impression** of being in three dimensions."

¹ *Savanna* refers to large grassy plains in the African continent.

² If something is *beleaguered*, it is in a very difficult situation.

This painting of a pair of humpback whales by Sophie Green is called *Concordia*.



H Green adds that hope is an embedded theme in her collections and is also the source of her collection's **ambiguous** title.

I “The name *impermanence* is open to **interpretation**—it kind of **implies** the impermanence of certain species and ecosystems,” she says. “But it could also represent the impermanence of our problems. There’s a dark side, but there can also be a light side.”

Adapted from “This artist’s animal paintings bridge a gap between photography and reality,” by Simon Ingram: National Geographic, October 2022

Simon Ingram is a writer for *National Geographic* and other well-known publications. He is also an editor and a photojournalist.

ART AS ACTIVISM

J It started as a countercultural art movement in 2001. After years studying at the Academy of Fine Arts, Kinshasa—following teachers’ advice on creating work with “proper” materials, such as resin and plaster of Paris—some students in the Democratic Republic of the Congo (DRC) decided to do something different. They created art with what was in their immediate environment, including tires, exhaust pipes, foam, plastic bottles, antennas, tins that had held milk or paint, feathers, CDs, rubber slippers, and other discarded items.

K This work, the artists believed, felt familiar to a Congolese audience and spoke to a particularly egregious³ aspect of Congolese life: waste. Waste generated locally by citizens. Waste dumped in the country by hyper-consumerist nations. Waste **triggered** by the endless extraction of resources from the DRC’s earth, or the rapacious⁴ collection of the same above land.

L In Kinshasa, gutters are brimming with nonrecyclable plastic bottles. Markets are awash with second- and third-hand goods, castoffs from high-income countries. In areas where international companies mine for cobalt—a precious component of smartphone batteries—frequent discharges contaminate river systems and surrounding life. By repurposing waste to create sculpture and performance art, the artists wanted to dial up the public’s acuity⁵ toward an **ongoing** emergency. Waste also provided the artists with an opening to comment on other fraught⁶ sociopolitical issues.

³If an action is **egregious**, it is wrong and offensive.

⁴Someone who is **rapacious** is very greedy.

⁵**Acuity** refers to the sharpness of one’s thoughts, vision, or hearing.

⁶If something is **fraught**, it is stressful or causes anxiety.



▲ **Robot Annonce**, a wearable sculpture by Jared Kalenga made using old radio parts



▲ A figure made from automobile parts by Precy Numbi

M *Robot Annonce*, for example, is a wearable sculpture by Jared Kalenga made of broken radio parts. It seeks to raise awareness about the ever-spreading reach of **fake** news. Precy Numbi's figure made out of automobile parts is a way of protesting the millions of "garbage cars" imported into Africa every year—secondhand vehicles that discourage the growth of the continent's own auto industry. And *Femme Électrique*, Falonne Mambu's creation made of electric wires, is double-edged. It speaks to the paucity of electric power service in the DRC and, simultaneously, what goes on in the dark: assaults and kidnappings. Mambu's inspiration for the work was drawn from periods in her life when she was homeless.

N These socially conscious creators who turn refuse into protest art "are out here pushing limits," says Yvon Edoumou, founder of the Galerie Malabo in Kinshasa. "We don't see a lot of that."

Adapted from "Transforming Trash Into Protest Art," by Ayodeji Rotinwa: National Geographic Magazine, June 2022

Ayodeji Rotinwa is a writer, editor, and journalist whose work has appeared in *National Geographic Magazine* and other well-known publications.

ART FOR SCIENCE

- o The art of origami has existed in Japan since at least the 17th century. Initially, origami models were simple and used largely for ceremonial purposes. But in the mid-20th century, origami master Akira Yoshizawa helped elevate paper folding to a fine art, breathing life and personality into each creature he designed.

- P** In the late 1950s, Yoshizawa's delicate forms inspired Tomoko Fuse, now one of the **foremost** origami artists in Japan. Her father gave her Yoshizawa's second origami book when she was recovering from diphtheria⁷ as a child. Fuse methodically crafted every model, and she's been entranced with origami ever since. "It's like magic," she says. "Just one flat paper becomes something wonderful."
- Q** Among her many achievements, Fuse is famous for her advances in modular⁸ origami, which uses interlocking units to create models with greater flexibility and potential complexity. But she thinks of her work as less about creation than about discovering something that's already there. She describes her process as if she's watching from afar, following wherever the paper leads her. "Suddenly, beautiful patterns come out."
- R** Indeed, origami taps into patterns that echo throughout the universe, seen in natural forms such as leaves emerging from a bud, or insects tucking their wings. For these **exquisite** folds to become scientifically useful, however, researchers must not only discover the patterns but also understand how they work. And that requires math.
- S** Putting numbers to origami's intriguing patterns has long driven the work of Thomas Hull, a mathematician at Western New England University in Springfield, Massachusetts. Hull still remembers unfolding a paper crane at age 10 and marveling at the ordered creases⁹ in the flat sheet. There are rules at play that allow this to work, he recalls thinking. Hull and others have spent decades working to understand the mathematics governing the world of origami.

⁷Diphtheria is a serious health condition that affects breathing and swallowing.

⁸If something is **modular**, it is made of components that can be easily attached or detached.

⁹Creases are the lines created by folding something.



Mathematicians do not fully understand the math behind the curving folds of this origami structure.



This golden disc—a light shield for telescopes in space—is designed to fold into a cylinder for launch.

T In his office are an array of models that are folded in intriguing shapes or move in unexpected ways. One is an impossible-looking sheet folded with ridges of concentric¹⁰ squares, which cause the paper to twist in an elegant swoop known as a hyperbolic paraboloid. Another is a sheet folded in a series of mountains and valleys called the Miura-ori pattern, which collapses or opens with a single tug. Dreamed up by astrophysicist Koryo Miura in the 1970s, the pattern was used to compact the solar panels of Japan's Space Flyer Unit, which launched in 1995.

¹⁰ Concentric shapes surround each other, and increase in size the further out they get.

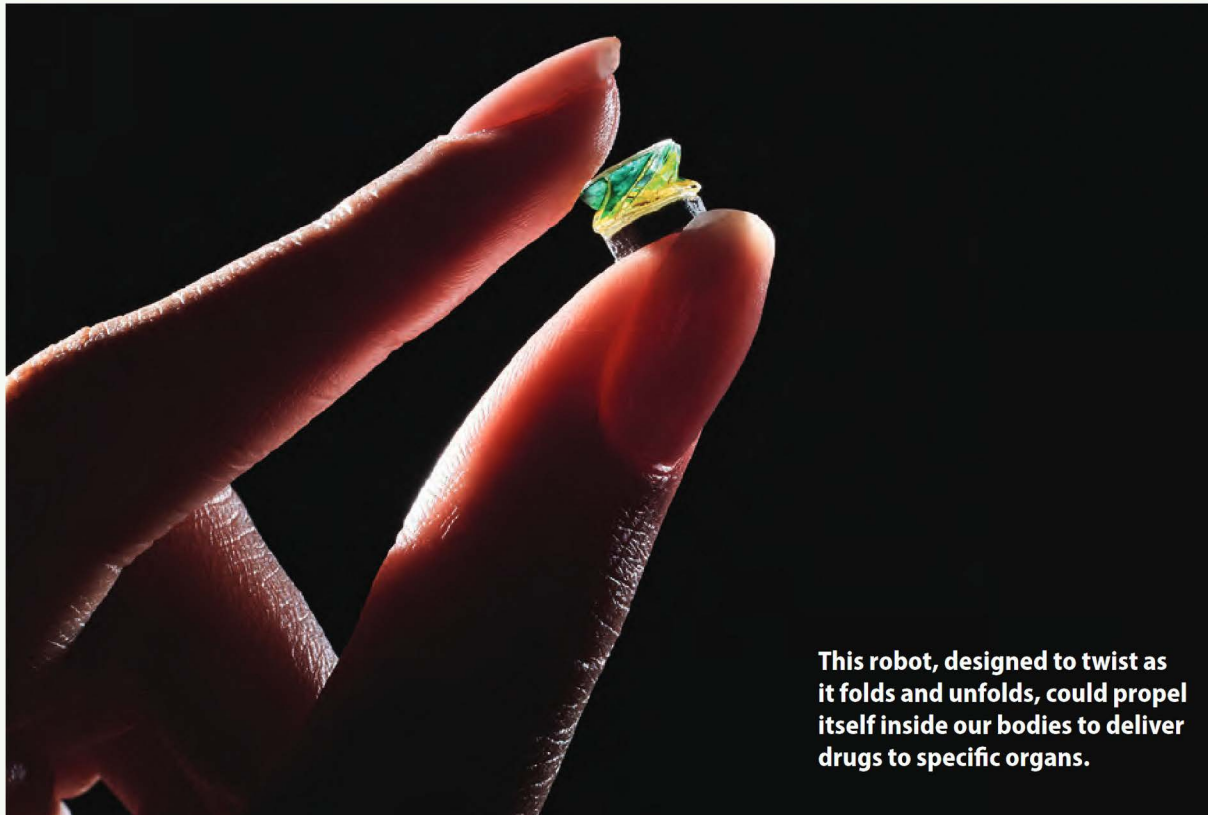
u Origami is now pushing the limits of what scientists think is possible, particularly at the tiniest of scales. At the University of Pennsylvania's Singh Center for Nanotechnology, Marc Miskin, an electrical engineer, has been crafting an army of robots, each no bigger than a speck of dust. Such small bots require big creativity. At tiny scales, forces like friction are enormous: gears don't turn, wheels don't spin, and belts don't run. That's where origami comes in. Fold patterns will bend and move the same way at any size, at least **theoretically**.

v Miskin sees a world of possible ways these tiny bots could be used, from manufacturing to medicine. And the venerable¹¹ art form of origami has provided him and other innovators with a new tool kit to ignite the imagination and create technologies once thought impossible.

Adapted from "The Future Is Folded," by Maya Wei-Haas: National Geographic Magazine, February 2023

Maya Wei-Haas is an award-winning science writer with a particular interest in rocks and reactions. She holds a Ph.D. in Earth Science from the Ohio State University, and is a staff writer for *National Geographic Magazine*.

¹¹ If something is venerable, it is highly revered or respected.



This robot, designed to twist as it folds and unfolds, could propel itself inside our bodies to deliver drugs to specific organs.

UNDERSTANDING THE READING

A UNDERSTAND MAIN IDEAS How do the art forms in the reading passage make an impact? Choose the correct answers.

1. Sophie Green's hyper-realistic paintings _____.
 - a. promote conservation by allowing people to connect with animals
 - b. educate people about animals by highlighting their amazing traits
2. The art made from waste material in the DRC _____.
 - a. raises the international profile of the country
 - b. raises awareness of urgent issues
3. The art of origami _____.
 - a. teaches scientists how plants and animals move in nature
 - b. provides scientists with a new way to design products

B UNDERSTAND DETAILS Match the parts to complete the descriptions.

- | | |
|-------------------------------|--|
| 1. ____ The Watcher | a. highlights the spread of misinformation in the media. |
| 2. ____ Impermanence | b. was made use of in a space vehicle. |
| 3. ____ Robot Annonce | c. has eyes that attract the viewer's gaze. |
| 4. ____ Femme Électrique | d. draws attention to a country's unreliable power supply. |
| 5. ____ The Miura-ori pattern | e. is a collection of 14 paintings of animals. |

C UNDERSTAND DETAILS Read the sentences. Choose **T** for true, **F** for false, or **NG** for not given.

- | | | | |
|---|----------|----------|-----------|
| 1. Sophie Green prides herself on her creativity. | T | F | NG |
| 2. Green prefers paints that dry quickly. | T | F | NG |
| 3. Most of the waste products in the DRC are from other nations. | T | F | NG |
| 4. Precy Numbi wants his country's auto industry to slow the rate of manufacturing. | T | F | NG |
| 5. At large scales, gears and wheels work better than fold patterns. | T | F | NG |

Critical Thinking

D SUMMARIZE Write down briefly what each person does or did. Paraphrase your answers.

1. Jared Kalenga _____.
2. Yvon Edoumou _____.
3. Akira Yoshizawa _____.
4. Tomoko Fuse _____.
5. Thomas Hull _____.
6. Koryo Miura _____.
7. Marc Miskin _____.